108-key QMK keyboard controller, matrix and RGB

Rev B. Connected circuit and footprint draft. Read status.md before fabrication.

RP2040 controller

File: controller.kicad_sch

Controller bypass and harness

File: power_io.kicad_sch

Matrix switches 1 to 18

File: matrix_1.kicad_sch

Matrix switches 19 to 36

File: matrix_2.kicad_sch

Matrix switches 37 to 54

File: matrix_3.kicad_sch

Matrix switches 55 to 72

File: matrix_4.kicad_sch

Matrix switches 73 to 90

File: matrix_5.kicad_sch

Matrix switches 91 to 108

File: matrix_6.kicad_sch

RGB level shifting

File: rgb_driver.kicad_sch

Key RGB 1 to 18

File: rgb_1.kicad_sch

Key RGB 19 to 36

File: rgb_2.kicad_sch

Key RGB 37 to 54

File: rgb_3.kicad_sch

Key RGB 55 to 72

File: rgb_4.kicad_sch

Key RGB 73 to 90

File: rgb_5.kicad_sch

Key RGB 91 to 108

File: rgb_6.kicad_sch

Sheet: /
File: keyboard.kicad_sch

Title:

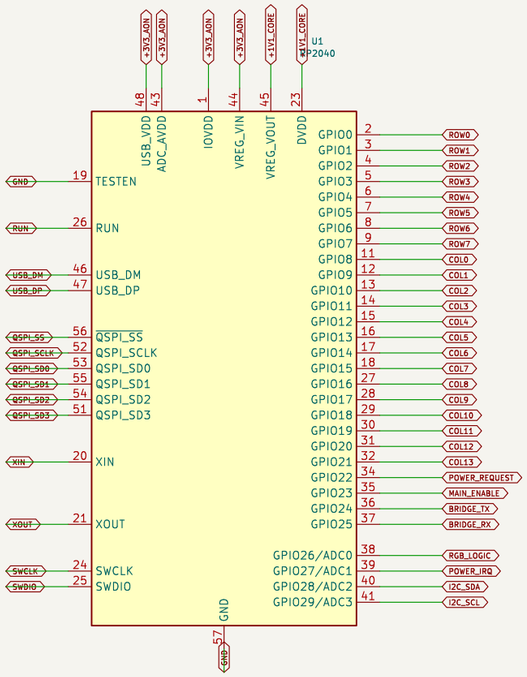
Size: A3 | Date:

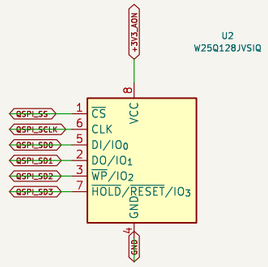
KiCad E.D.A. 9.0.7

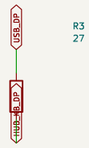
Rev:

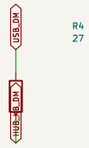
Id: 16/16

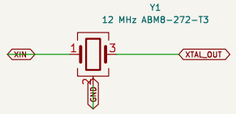
RP2040 + 16 MB QSPI flash. External regulated 3.3 V and 5 V inputs.
USB connects to an internal hub downstream port. UART bridge firmware remains to be integrated.





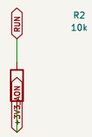


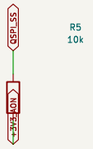


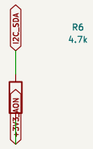


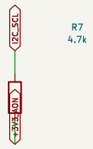


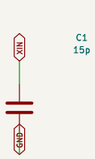


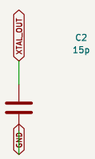


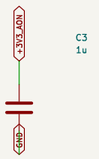


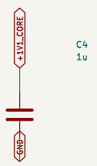




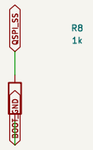














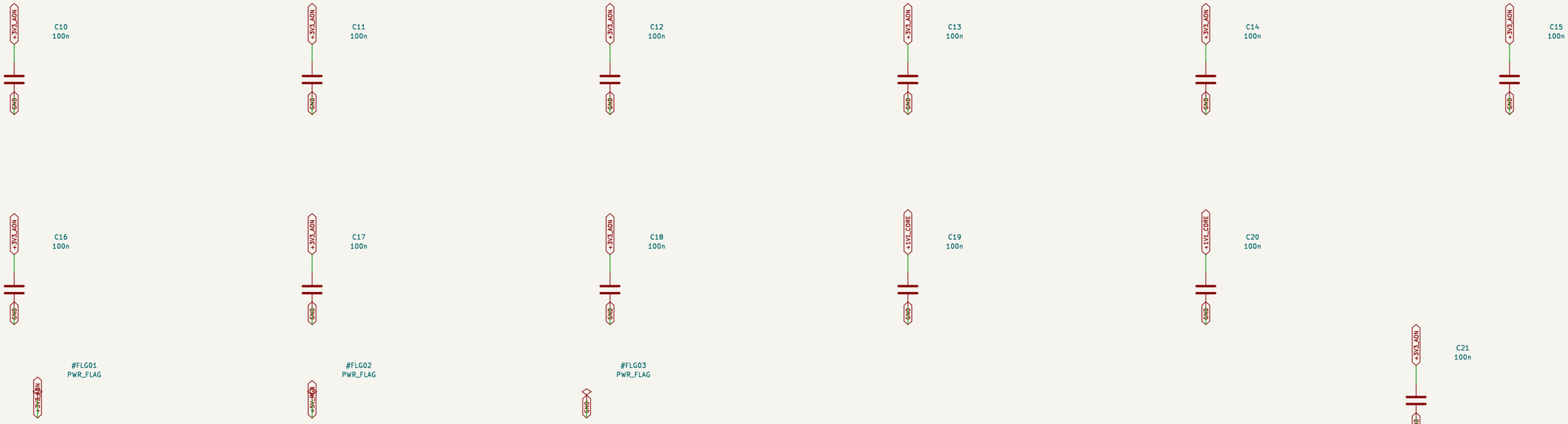
Connected schematic; hardware and manufacturing validation pending

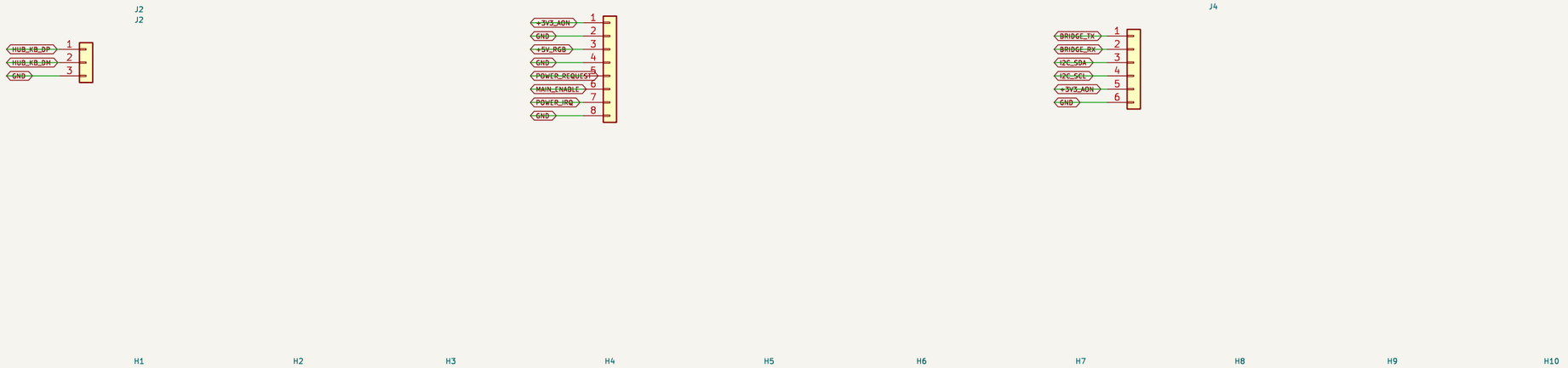
Sheet: /RP2040 controller/
File: controller.kicad_sch

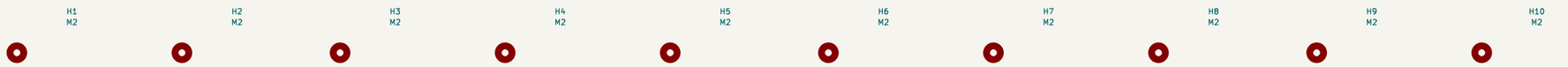
Title: RP2040 controller

Size: A3	Date:	**Rev: B — engineering draft**
KiCad E.D.A. 9.0.7		Id: 2/16

J2: USB data pair and ground only. J3/J4: power and low-speed control harness.
Regulated rails originate on the future power board. No battery may connect directly here.







Connected schematic; hardware and manufacturing validation pending

Sheet: /Controller bypass and harness/
File: power_io.kicad_sch

Title: Controller bypass and harness

Size: A3	Date:	**Rev: B — engineering draft**
KiCad E.D.A. 9.0.7		Id: 3/16

COL2ROW: switch connects column to diode anode. Diode cathode (pin 1) connects row.
19.05 mm physical pitch. Actual matrix assignments are in key-matrix.csv.









































































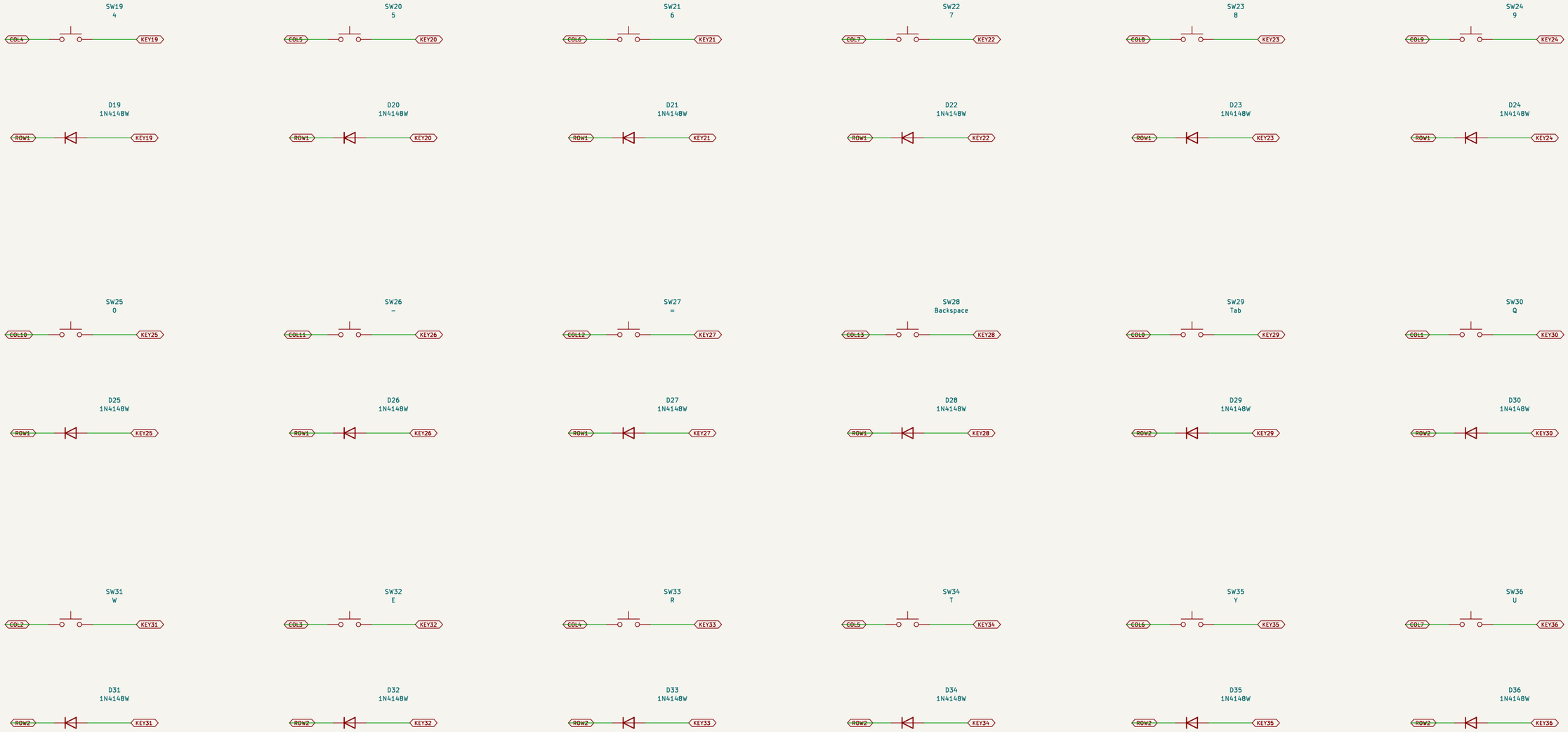
Connected schematic; hardware and manufacturing validation pending

Sheet: /Matrix switches 1 to 18/
File: matrix_1.kicad_sch

Title: Matrix switches 1 to 18

Size: A3	Date:	**Rev: B — engineering draft**
KiCad E.D.A. 9.0.7		Id: 4/16

COL2ROW: switch connects column to diode anode. Diode cathode (pin 1) connects row.
19.05 mm physical pitch. Actual matrix assignments are in key-matrix.csv.



Connected schematic; hardware and manufacturing validation pending

Sheet: /Matrix switches 19 to 36/
File: matrix_2.kicad_sch

Title: Matrix switches 19 to 36

Size: A3	Date:	**Rev: B — engineering draft**
KiCad E.D.A. 9.0.7		Id: 5/16

COL2ROW: switch connects column to diode anode. Diode cathode (pin 1) connects row.
19.05 mm physical pitch. Actual matrix assignments are in key-matrix.csv.









































































Connected schematic; hardware and manufacturing validation pending

Sheet: /Matrix switches 37 to 54/
File: matrix_3.kicad_sch

Title: Matrix switches 37 to 54

Size: A3	Date:	**Rev: B — engineering draft**
KiCad E.D.A. 9.0.7		Id: 6/16

COL2ROW: switch connects column to diode anode. Diode cathode (pin 1) connects row.
19.05 mm physical pitch. Actual matrix assignments are in key-matrix.csv.









































































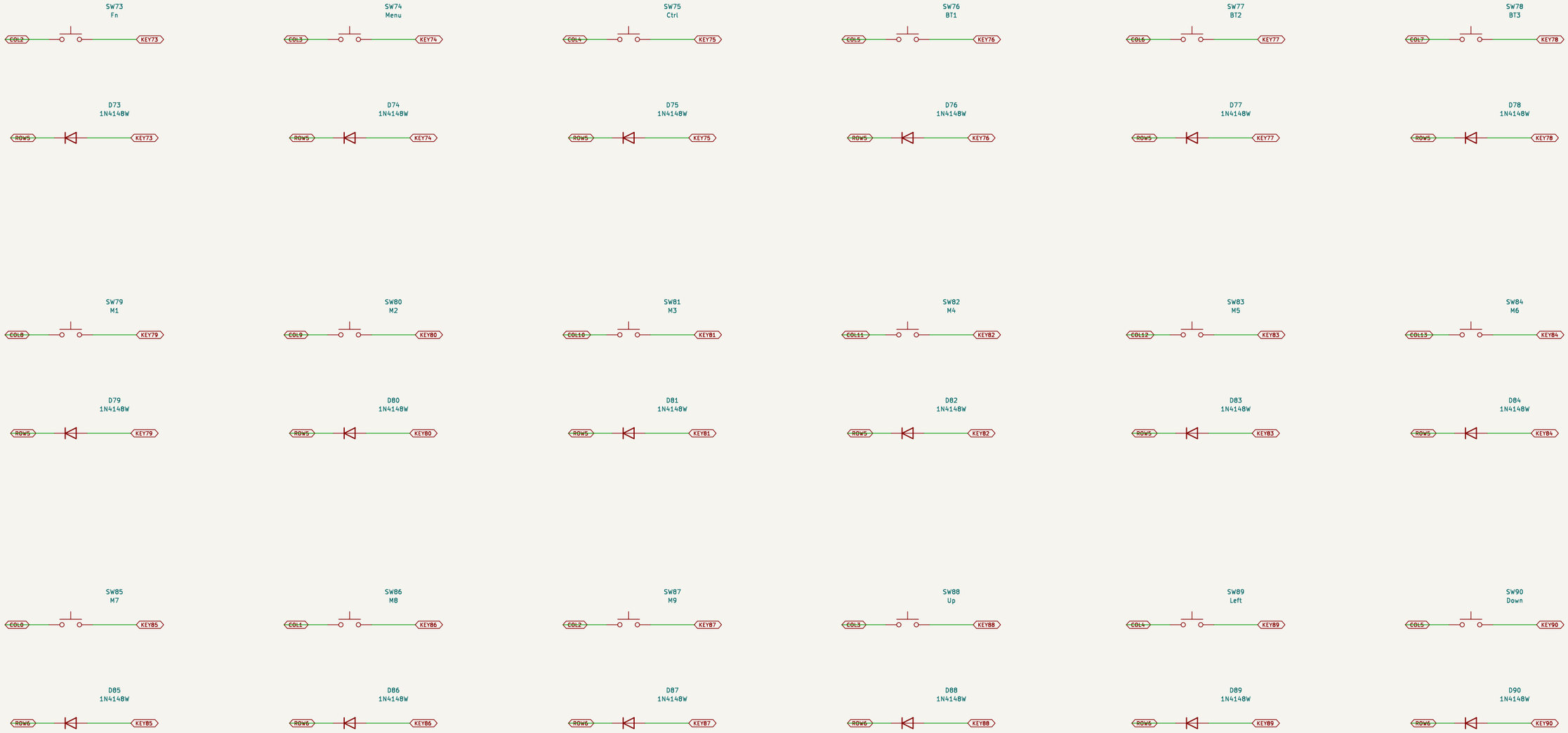
Connected schematic; hardware and manufacturing validation pending

Sheet: /Matrix switches 55 to 72/
File: matrix_4.kicad_sch

Title: Matrix switches 55 to 72

Size: A3 | Date: | **Rev: B — engineering draft**
KiCad E.D.A. 9.0.7 | Id: 7/16

COL2ROW: switch connects column to diode anode. Diode cathode (pin 1) connects row.
19.05 mm physical pitch. Actual matrix assignments are in key-matrix.csv.



Connected schematic; hardware and manufacturing validation pending

Sheet: /Matrix switches 73 to 90/
File: matrix_5.kicad_sch

Title: Matrix switches 73 to 90

Size: A3	Date:	**Rev: B — engineering draft**
KiCad E.D.A. 9.0.7		Id: 8/16

COL2ROW: switch connects column to diode anode. Diode cathode (pin 1) connects row.
19.05 mm physical pitch. Actual matrix assignments are in key-matrix.csv.









































































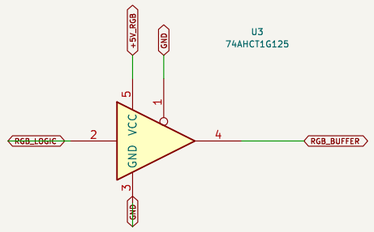
Connected schematic; hardware and manufacturing validation pending

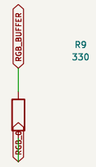
Sheet: /Matrix switches 91 to 108/
File: matrix_6.kicad_sch

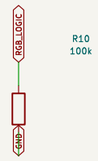
Title: Matrix switches 91 to 108

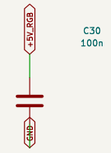
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KiCad E.D.A. 9.0.7 | Id: 9/16

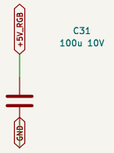
109 RGB pixels: 108 keys followed by the battery indicator.
Keep the battery pixel separate in software. Hardware power limiting is required on the supply board.

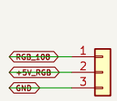












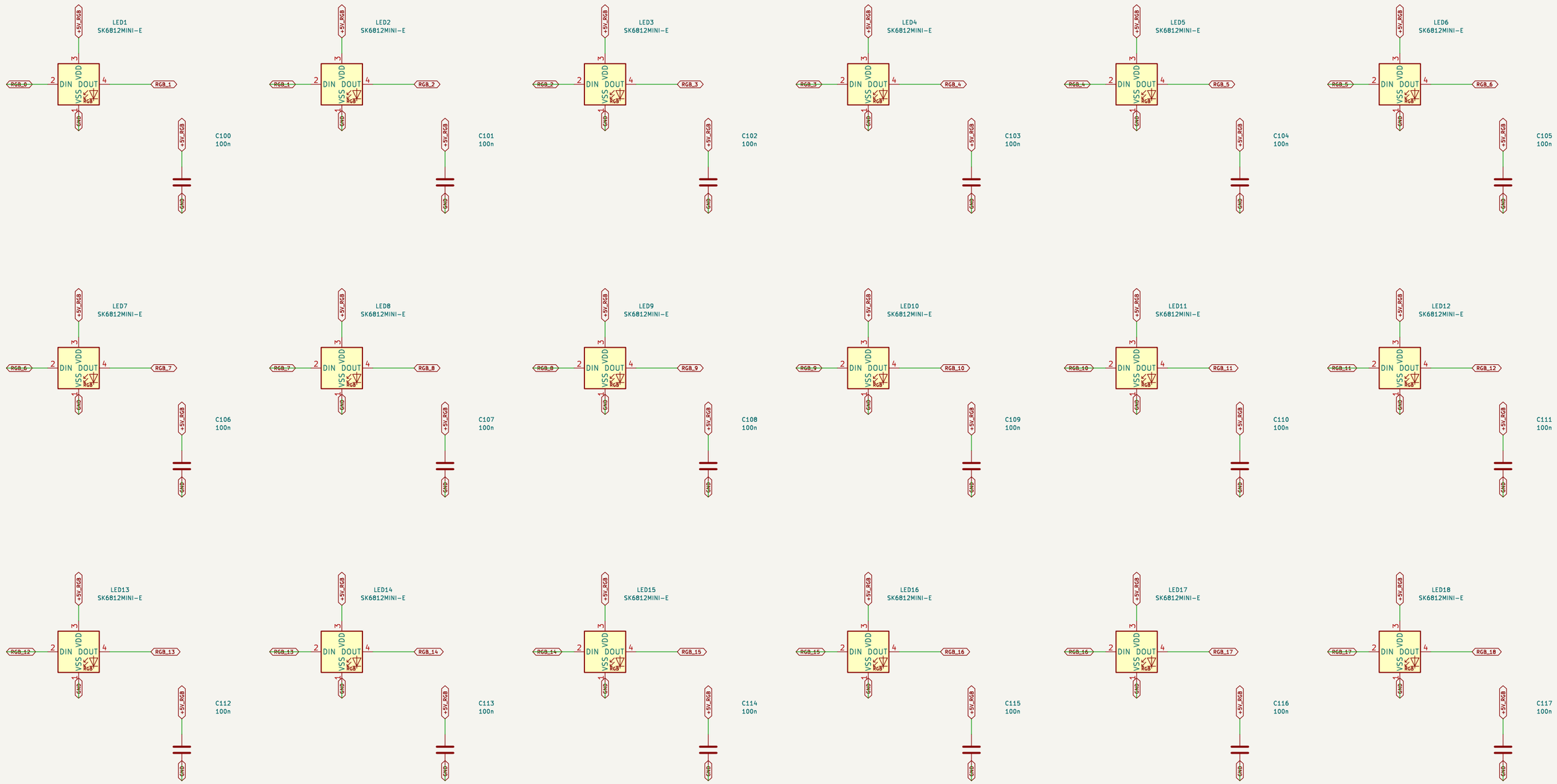
Connected schematic; hardware and manufacturing validation pending

Sheet: /RGB level shifting/
File: rgb_driver.kicad_sch

Title: RGB level shifting

Size: A3	Date:	**Rev: B – engineering draft**
KiCad E.D.A. 9.0.7		Id: 10/16

SK6812MINI-E reverse-mounted underneath PCB; light shines up through routed opening.
Pins per OPSCO C5149201: 1 GND, 2 DIN, 3 VDD, 4 DOUT. GRB data order. 100 nF per pixel.



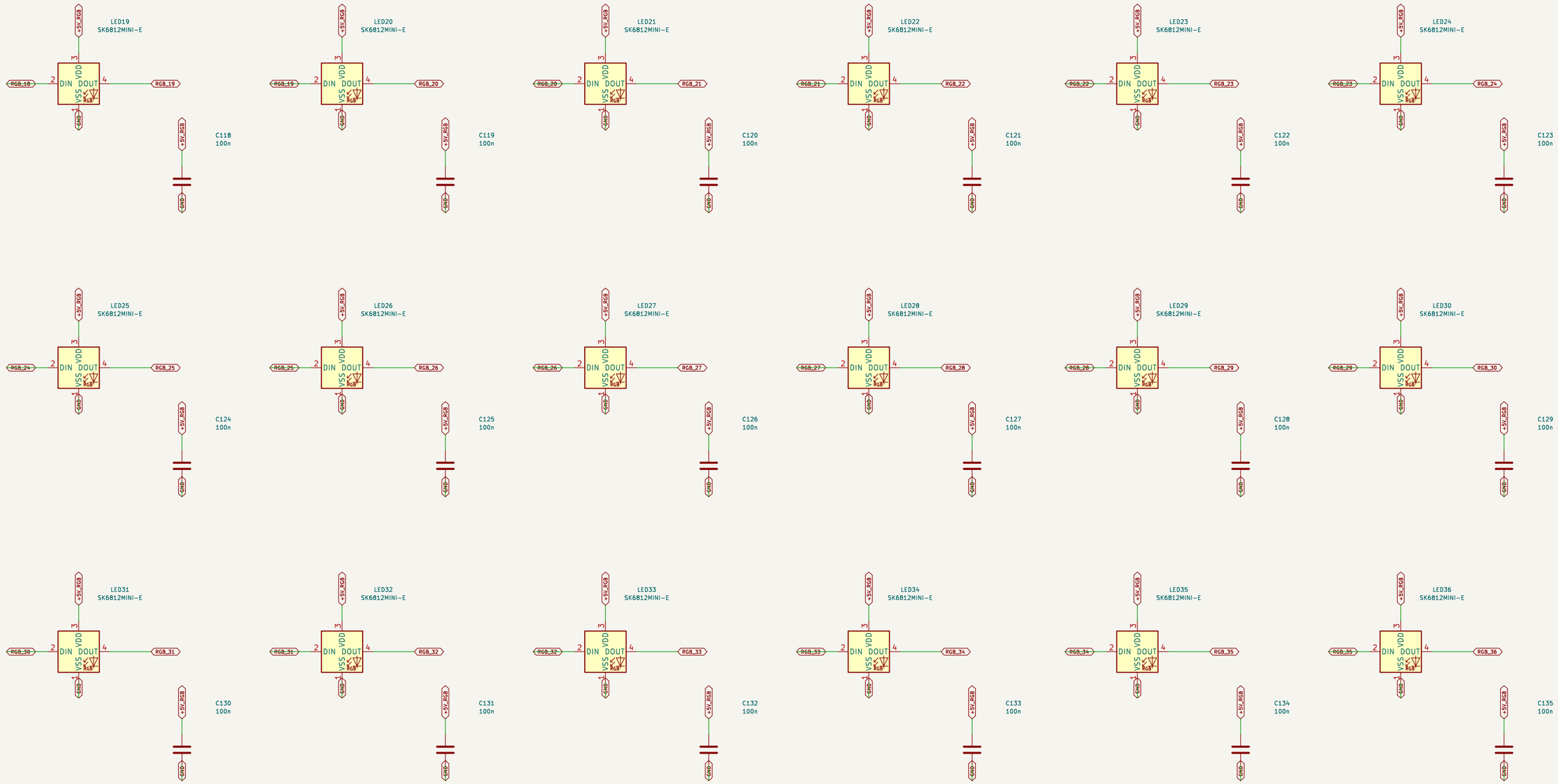
Connected schematic; hardware and manufacturing validation pending

Sheet: /Key RGB 1 to 18/
File: rgb_1.kicad_sch

Title: Key RGB 1 to 18

Size: A3	Date:	**Rev: B – engineering draft**
KiCad E.D.A. 9.0.7		Id: 11/16

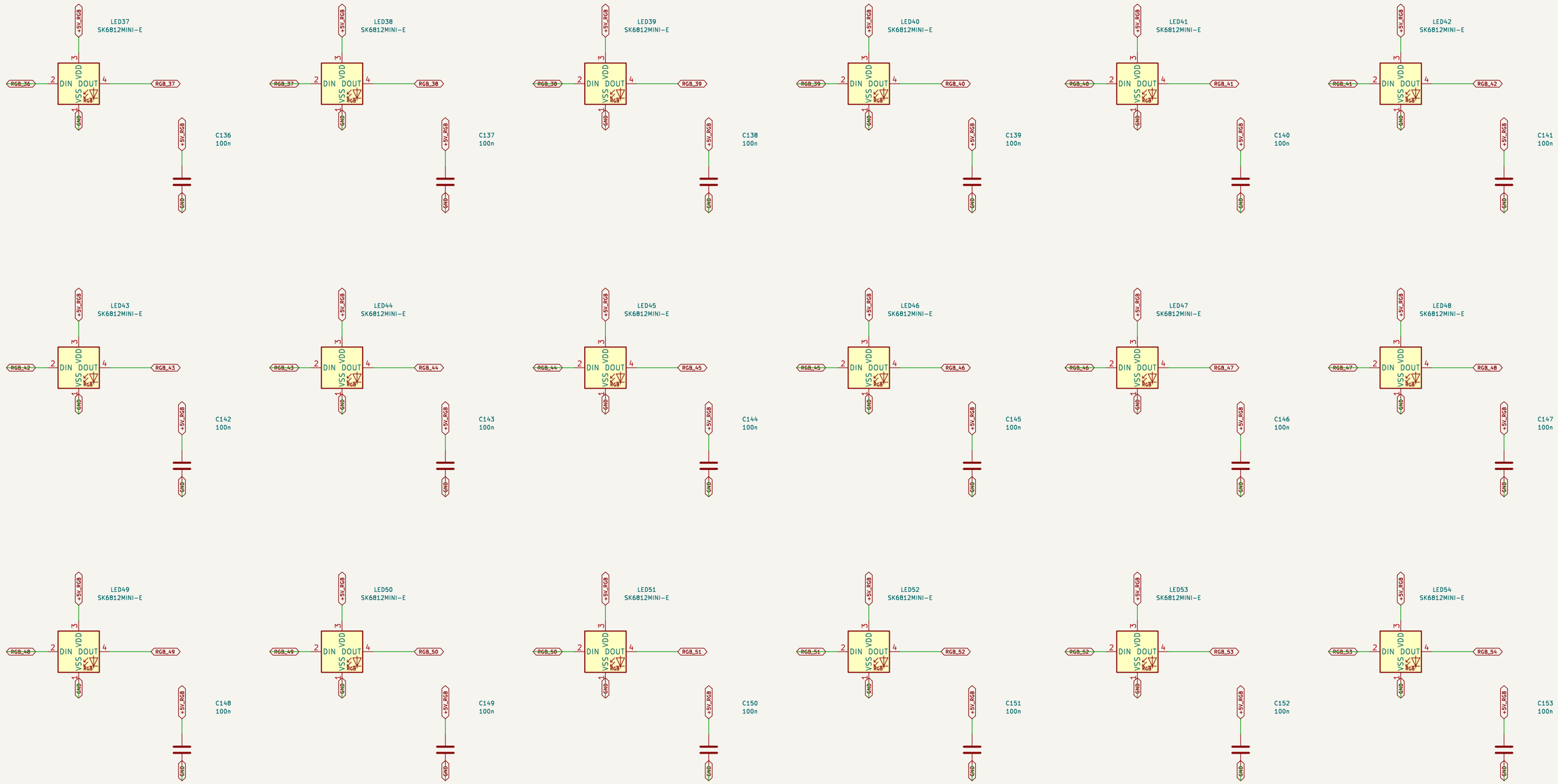
SK6812MINI-E reverse-mounted underneath PCB; light shines up through routed opening.
Pins per OPSCO C5149201: 1 GND, 2 DIN, 3 VDD, 4 DOUT. GRB data order. 100 nF per pixel.



Connected schematic; hardware and manufacturing validation pending

Sheet: /Key RGB 19 to 36/ File: rgb_2.kicad_sch		
Title: Key RGB 19 to 36		
Size: A3	Date:	**Rev: B – engineering draft**
KiCad E.D.A. 9.0.7		Id: 12/16

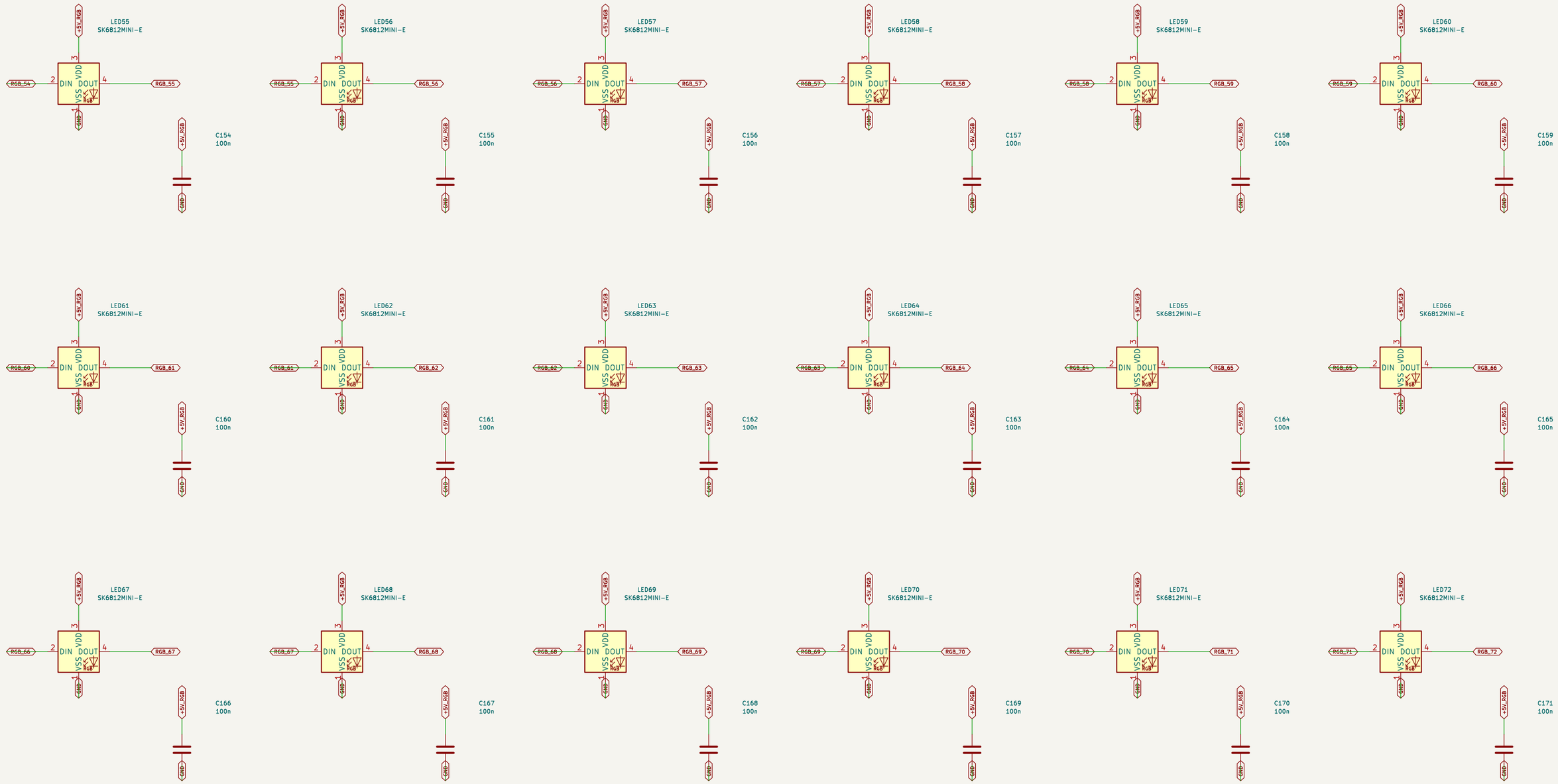
SK6812MINI-E reverse-mounted underneath PCB; light shines up through routed opening.
Pins per OPSCO C5149201: 1 GND, 2 DIN, 3 VDD, 4 DOUT. GRB data order. 100 nF per pixel.



Connected schematic; hardware and manufacturing validation pending

Sheet: /Key RGB 37 to 54/		
File: rgb_3.kicad_sch		
Title: Key RGB 37 to 54		
Size: A3	Date:	**Rev: B – engineering draft**
KiCad E.D.A. 9.0.7		Id: 13/16

SK6812MINI-E reverse-mounted underneath PCB; light shines up through routed opening.
Pins per OPSCO C5149201: 1 GND, 2 DIN, 3 VDD, 4 DOUT. GRB data order. 100 nF per pixel.



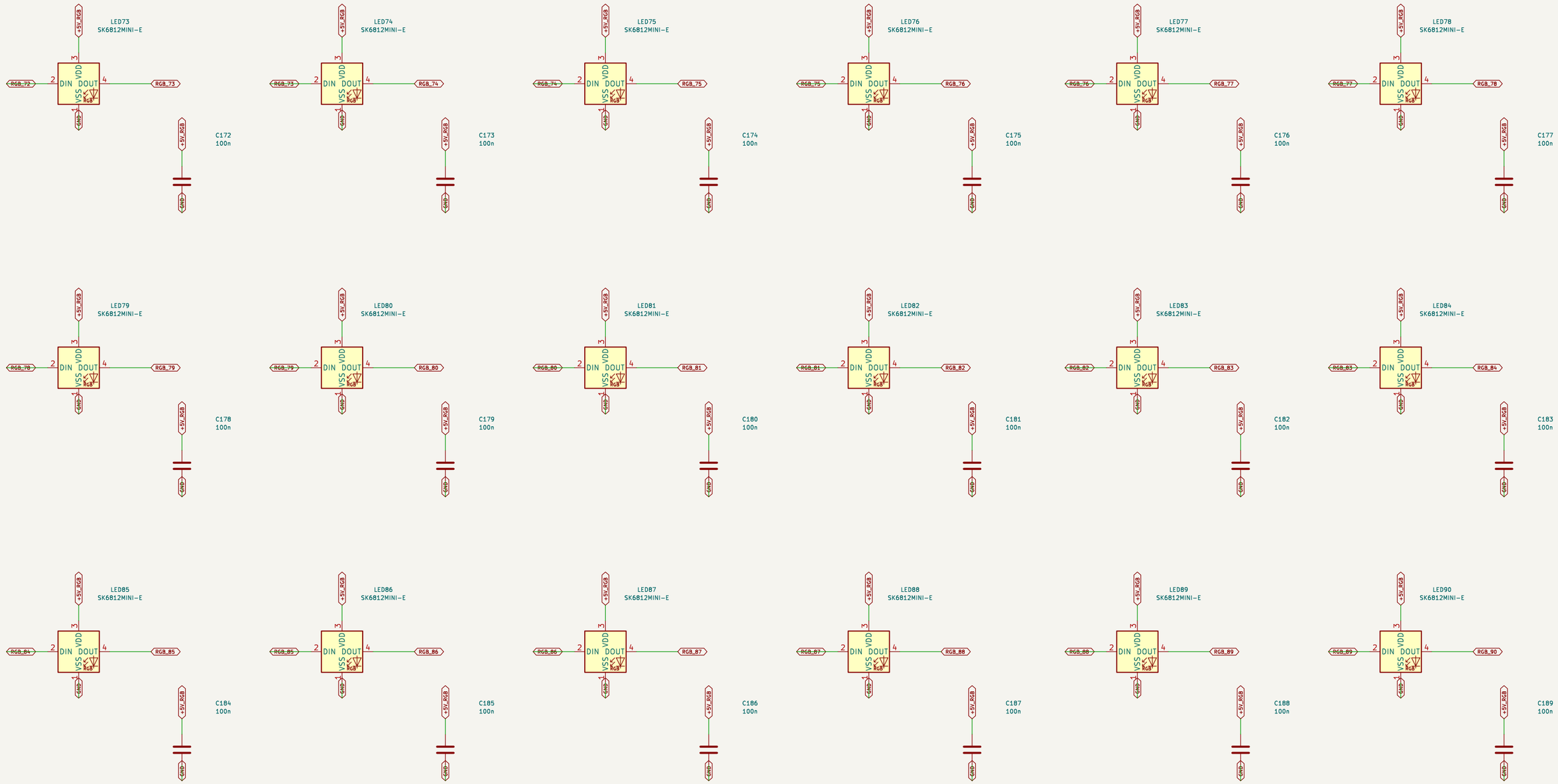
Connected schematic; hardware and manufacturing validation pending

Sheet: /Key RGB 55 to 72/
File: rgb_4.kicad_sch

Title: Key RGB 55 to 72

Size: A3	Date:	**Rev: B – engineering draft**
KiCad E.D.A. 9.0.7		Id: 14/16

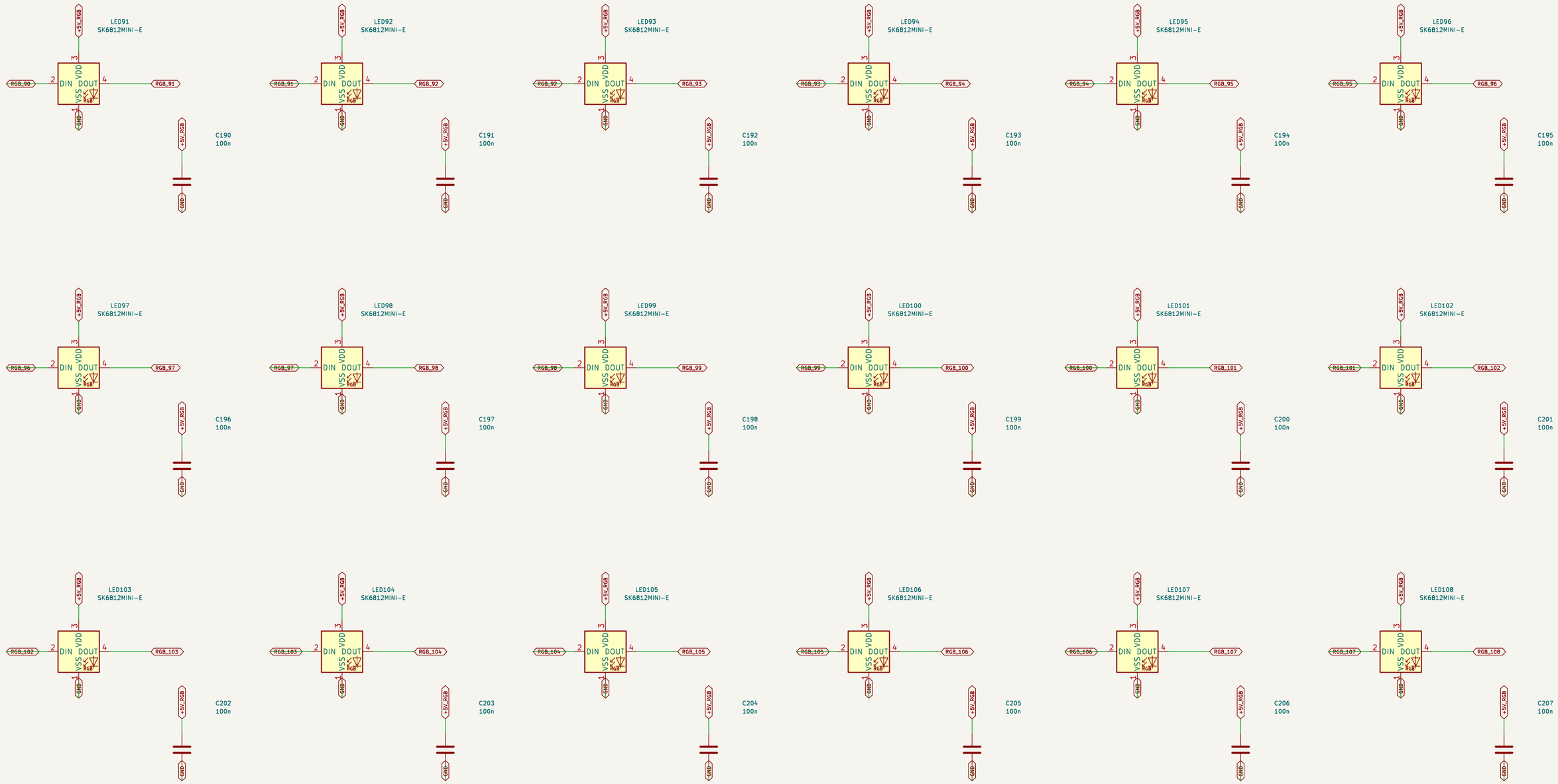
SK6812MINI-E reverse-mounted underneath PCB; light shines up through routed opening.
Pins per OPSCO C5149201: 1 GND, 2 DIN, 3 VDD, 4 DOUT. GRB data order. 100 nF per pixel.



Connected schematic; hardware and manufacturing validation pending

Sheet: /Key RGB 73 to 90/ File: rgb_5.kicad_sch		
Title: Key RGB 73 to 90		
Size: A3	Date:	**Rev: B – engineering draft**
KiCad E.D.A. 9.0.7		Id: 15/16

SK6812MINI-E reverse-mounted underneath PCB; light shines up through routed opening.
Pins per OPSCO C5149201: 1 GND, 2 DIN, 3 VDD, 4 DOUT. GRB data order. 100 nF per pixel.



Connected schematic; hardware and manufacturing validation pending

Sheet: /Key RGB 91 to 108/ File: rgb_6.kicad_sch		
Title: Key RGB 91 to 108		
Size: A3	Date:	**Rev: B – engineering draft**
KiCad E.D.A. 9.0.7		Id: 16/16